



UNIVERSITY OF
ILLINOIS CHICAGO

DEPARTMENT OF CHEMICAL ENGINEERING

TRANSPORT PHENOMENA I

CHE 311, SPRING 2026



Department of Chemical Engineering
College of Engineering
929 W. Taylor St., 226 EIB (MC 110)
Chicago, IL 60607

Phone 312.996.3424
Fax 312.996.0808
Email cheuic@uic.edu
Web che.uic.edu

COLLEGE OF ENGINEERING, UIC

TRANSPORT PHENOMENA I

CHE 311, SPRING 2026

I. INSTRUCTOR & COURSE DETAILS

INSTRUCTOR

KOROSH TORABI

Email address: korosh@uic.edu

Drop-In Office Hours: Wednesdays 11 AM – 12 PM, EIB 246

TEACHING ASSISTANT

SHAHRIAR NAMVAR

E-mail: snamva2@uic.edu

Drop-In Office Hours: Fridays 11 AM – 12 PM, EIB 139

COURSE BLACKBOARD SITE

<https://uic.blackboard.com>

Students are expected to log in to the course site regularly to stay informed about course developments, upload assignments, and communicate with classmates. Email the Learning Technology Solutions team at LTS@uic.edu for all technical questions about Blackboard.

COURSE MODALITY AND SCHEDULE

This course is taught on campus.

Mondays and Wednesdays 2:30 PM - 3:45 PM (3 credit hours), EIB Room 124

II. COURSE INFORMATION

CATALOG COURSE DESCRIPTION AND PREREQUISITE/COREQUISITE STATEMENT

Momentum transport phenomena in chemical engineering. Fluid statics. Fluid mechanics; laminar and turbulent flow; boundary layers; flow over immersed bodies.

Prerequisite(s): Credit or concurrent registration in CHE 210; and MATH 220; and CHE 205.

BRIEF LIST OF TOPICS TO BE COVERED

Mathematics review (vector and tensor Algebra and calculus), momentum flux, Newton's law of viscosity, shell momentum balances (unidirectional flow problems), equations of change, two-dimensional flow, turbulent flow, and friction factors and their applications.

COURSE RATIONALE AND GROWTH MINDSET

This course presents the fundamental physical concepts related to *fluid dynamics* and *momentum transport*. Knowledge of these topics is needed in subsequent undergraduate chemical engineering courses, specifically energy and mass transport (CHE 312). Essentially, this course serves as a bridge between basic physics and

chemical engineering; while the subjects covered are rooted in classical physics, the applications are specifically tailored for chemical engineers.

It is assumed that students in the course have a thorough grounding in elementary mathematics, and this mathematical background will be utilized; some review of mathematics may be necessary at various points in the course.

It is strongly suggested that you read the textbook carefully, do the homework problems in detail without consulting other solutions, ask thoughtful questions in class and during office hours, discuss the material with your classmates, and work on additional problems to master it.

Course materials and assignments can be complex and challenging, but they are crucial to intellectual growth and development. There are times when you may need extra help. Students who attend class consistently, complete all assignments, thoughtfully engage with feedback on work, develop good study strategies, visit the TA, and contact faculty when struggling can develop a thorough understanding of the course material and ultimately succeed in the course!

COURSES LEARNING OUTCOMES

At the completion of this course, students should be able to:

- Use mathematical language to perform vector and tensor operations relevant to transport phenomena.
- Comprehend the fundamental physical principles of momentum transport and apply the principle of conservation of momentum to solve basic fluid dynamics problems.
- Set up the differential momentum balance and specify its boundary conditions for various flow problems.
- Solve ordinary differential equations and apply boundary conditions to determine steady-state one-dimensional flow velocity and stress profiles.
- Solve partial differential equations using the methods of separation of variables and combinations of variables to determine the velocity profiles of flow fields with two independent variables.
- Comprehend the physical significance and mathematical formulation of empirical expressions for turbulent momentum transport.
- Understand the definition of friction factors to solve various flow problems.

ABET STUDENT LEARNING OUTCOMES

- ABET Student Outcome 1: an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
- ABET Student Outcome 2: an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
- ABET Student Outcome 7: an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

REQUIRED TEXTBOOK

Transport Phenomena 2nd Edition

by R. Byron Bird, Warren E. Stewart, Edwin N. Lightfoot

Note: Revised 2nd edition is preferred.

III. COURSE POLICIES & CLASSROOM EXPECTATIONS

GRADING POLICY AND POINT BREAKDOWN

- Homework: 10%
- Quizzes: 20%
- Midterm Exam I: 20%
- Midterm Exam II: 20%
- Final Exam 30%

ASSIGNMENTS

- **Reading assignments:** Assigned from the textbook. No submission is required.
- **Problem sets (Homework):** submit an electronic copy of your homework by the specified due date.

POLICY FOR MISSED OR LATE ASSIGNMENT OR EXAM

- **Homework:** Late submissions will not be considered for partial credit. Unreadable and unorganized homework will not be graded.
- **Quizzes:** No makeup will be provided for missed quizzes; however, the lowest quiz grade will be dropped from the final grade calculation.
- **Exams:** No makeup will be provided for missed exams. Students who need to miss an exam for any reason are expected to contact the instructor, preferably in advance of the exam date. With the appropriate documentation, the grade for the missed exam will be calculated based on the average of the other exam grades.

ATTENDANCE AND PARTICIPATION POLICY

Please email me if you face an unexpected situation that may impede your attendance and participation in required class and exam sessions.

EMAIL EXPECTATIONS

Students are responsible for all information instructors send to their UIC email and Blackboard accounts. Faculty messages should be regularly monitored and read in a timely fashion.

ACADEMIC INTEGRITY

As a student and member of the UIC community, you are expected to adhere to the [Community Standards of academic integrity](#), accountability, and respect. Please review the [UIC Student Disciplinary Policy](#) for additional information.

OTHER COURSE POLICIES

Regrading of exams must be requested within 2 days after receiving the grades. For any homework issues, please reach out to the TA.

IV. COURSE SCHEDULE

Week	Topics	Book Chapter
1	Conservation laws, flux, and balance equations Math Review	Chapter 0, Appendix A and Math Handout
2	Continuum fluid mechanics and Stress tensor Math review	Appendix A and Math Handout
3	Newton's law of viscosity and the combined momentum flux Math review	Chapter 1
4-5	Introduction to shell momentum balance Flow of a falling film	Chapter 2
6	Flow through a circular tube and modified pressure Creeping flow around a sphere	Chapter 2
7	Midterm Exam 1	Appendix A and Math Handout Chapters 0, 1 and 2
7	The equation of continuity and the equations of motion Drifting time derivative and Reynolds transport theorem	Chapter 3
7-8	Solving flow problems using the equations of change	Chapter 3
9-10	Unsteady flow and the method of combination of variables Unsteady flow and method of separation of variables	Chapter 4
11	(March 23-27) Spring break. No classes.	
12	Midterm Exam 2	Chapters 3 and 4
12	Introduction to turbulent flow and time-smoothed equations of change	Chapter 5
13-14	Friction factors for flow in tubes	Chapter 6
14	Friction factors for flow around submerged objects Pressure drop in packed beds	Chapter 6
15	Macroscopic momentum and mechanical energy balance	Chapter 7
16	Review and problem-solving	
17	Final Examination	Comprehensive

- Quizzes will be scheduled during regular class time and location with at least two days' notice.
- Midterm exams will be scheduled during regular class time and location.
- The final exam will be held on the date and time specified by the university's Registrar's Office.

DISCLAIMER

This syllabus is intended to provide students with guidance on what may be covered during the semester and will be followed as closely as possible. However, as the instructor, I reserve the right to modify, supplement, and make changes as needed to address course requirements. I will communicate such changes in advance through in-class announcements and in writing via Blackboard Announcements.

V. ACCOMMODATIONS

DISABILITY ACCOMMODATION PROCEDURES

UIC is committed to full inclusion and participation of people with disabilities in all aspects of university life. If you face or anticipate disability-related barriers while at UIC, please connect with the Disability Resource Center (DRC) at drc@uic.edu, via email at drc@uic.edu, or call (312) 413-2183 to create a plan for reasonable accommodations. To receive accommodation, you will need to disclose the disability to the DRC, complete an interactive registration process with the DRC, and provide me with a Letter of Accommodation (LOA). Upon receipt of an LOA, I will gladly work with you and the DRC to implement approved accommodations.

RELIGIOUS ACCOMMODATIONS

Following [campus policy](#), if you wish to observe religious holidays, you must notify me by the tenth day of the semester. If the religious holiday is observed on or before the tenth day of the semester, you must notify me at least five days before you will be absent. Please submit [this form](#) by email with the subject heading: "YOUR NAME: Requesting Religious Accommodation."

VI. CLASSROOM ENVIRONMENT

INCLUSIVE COMMUNITY

UIC values diversity and inclusion. Regardless of age, disability, ethnicity, race, gender, gender identity, sexual orientation, socioeconomic status, geographic background, religion, political ideology, language, or culture, we expect all members of this class to contribute to a respectful, welcoming, and inclusive environment for every other member of our class. If aspects of this course result in barriers to your inclusion, engagement, accurate assessment, or achievement, please notify me as soon as possible.

NAME AND PRONOUN USE

If your name does not match the name on my class roster, please let me know as soon as possible. My pronouns are he/him. I welcome your pronouns if you would like to share them with me. For more information about pronouns, see this page: <https://www.mypronouns.org/what-and-why>.

COMMUNITY AGREEMENT/CLASSROOM CONDUCT POLICY

- Be present by turning off cell phones and removing yourself from other distractions.
- Be respectful of the learning space and community. For example, no side conversations or unnecessary disruptions.
- Use preferred names and gender pronouns.
- Assume goodwill in all interactions, even in disagreement.
- Facilitate dialogue and value the free and safe exchange of ideas.
- Try not to make assumptions, have an open mind, seek to understand, and do not judge.
- Approach discussion, challenges, and different perspectives as an opportunity to "think out loud," learn something new, and understand the concepts or experiences that guide other people's thinking.
- Debate concepts, not the person.
- Be gracious and open to change when your ideas, arguments, or positions do not work or are proven wrong.
- Be willing to work together and share helpful study strategies.

CONTENT NOTICES AND TRIGGER WARNINGS

Our classroom provides an open space for a critical and civil exchange of ideas, inclusive of a variety of perspectives and positions. Some readings and other content may expose you to ideas, subjects, or views that may challenge you, cause you discomfort, or recall past negative experiences or traumas. I intend to discuss all subjects with dignity and humanity, as well as with rigor and respect for scholarly inquiry. If you would like me to be aware of a specific topic of concern, please email or visit my Student Drop-In Hours.

VII. RESOURCES: ACADEMIC SUCCESS, WELLNESS, AND SAFETY

We all need the help and the support of our UIC community. Please visit my office hours for course consultation and other academic or research topics. For additional assistance, please contact your assigned college advisor and visit the support services available to all UIC students.

ACADEMIC SUCCESS

- UIC [Tutoring Resources](#)
- College of Engineering [tutoring program](#)
- [Equity and Inclusion in Engineering Program](#)
- [UIC Library](#) and [UIC Library Research Guides](#)
- [Offices](#) supporting the UIC Undergraduate Experience and Academic Programs.
- [Student Guide for Information Technology](#)

WELLNESS

- Counseling Services: You may seek free and confidential services from the Counseling Center at <https://counseling.uic.edu/>.
- Access [U&I Care Program](#) for assistance with personal hardships.
- Campus Advocacy Network: Under Title IX, you have the right to an education free from any form of gender-based violence or discrimination. To make a report, email TitleIX@uic.edu. For more information or confidential victim services and advocacy, visit UIC's Campus Advocacy Network at <http://can.uic.edu/>.

SAFETY

- [UIC Safe App](#)—PLEASE DOWNLOAD FOR YOUR SAFETY!
- [UIC Safety Tips and Resources](#)
- [Night Ride](#)
- [Emergency Communications](#): By dialing 5-5555 from a campus phone, you can summon the Police or Fire for any on-campus emergency. You may also set up the complete number, (312) 355-5555, on speed dial on your cell phone.